Discovering Plate Approach Patterns Among Major League Baseball Batters Through Data Mining

Christopher Harris
College of Computing and Software Engineering
Kennesaw State University
Marietta, Georgia
charr396@students.kennesaw.edu

***Abstract*—This project examines patterns in Major League Baseball (MLB) batting behavior by analyzing plate approach statistics from qualified batters over the 2021-2025 seasons. The analysis uses the Baseball Savant MLB leaderboard, a publicly available dataset, and focuses on discovering natural sets of plate approaches from player-season observations based on the data in the leaderboard. This data measures plate discipline through swing and contact rates and includes observed results and expected results to view their effectiveness. By cross-comparing this data between the identified groups, it will be seen how each one relates to performance and outcome variance, focusing on pure pattern discovery.**

I. Introduction

The purpose of this project is to discover patterns in MLB batter data, revealing the different approaches and strategies players have when at the plate. This data can reveal the underlying discipline a player has, along with their traits. Are they an aggressive or passive hitter? Do they make consistent contact when they swing? These types of metrics determine where each player excels, identifying their skill and style type, no matter how successful they are, which can be useful for evaluating and comparing them on a comprehensive level.

II. Dataset Description

A. Data Source

The dataset used is derived from the Baseball Savant MLB leaderboard, and the data was exported through the platform's built-in CSV download functionality. The dataset displays performance data among qualified MLB batters over the past five seasons (2021-2025), defined as players with at least 500 plate appearances in a single season.

B. Data Size

In the selected dataset, there are 669 rows, each representing a player-season observation, and 11 columns representing plate approach measurements and outcome statistics.

C. Data Representation

The dataset describes individual plate approach behavior and tendencies, as well as outcome quality for an overall season. Plate approach statistics track how often a batter swings at a pitch, where they swing, and how frequently they make contact with the ball. These measurements relate to result statistics, summarized by the weighted on-base average (wOBA), signifying the observed outcomes of each plate approach. Expected statistics (xwOBA) are included to measure underlying performance based on the quality of contact made, rather than observed results, helping to exclude outcome variance due to luck. The data only accounts for a player's overall performance across an entire season, not a game-by-game analysis.

D. Key Measurements

Observation Data

- Player
- Season

Plate Approach Data

- K% - Strikeout Rate
- BB% - Walk Rate
- Z-Swing% - In-Zone Swing Rate
- O-Swing% - Out-of-Zone Swing Rate
- Z-Contact% - In-Zone Contact Rate
- O-Contact% - Out-of-Zone Contact Rate
- Swing% - Swing Rate

Performance Data

- wOBA - Weighted On-Base Average
- xwOBA - Expected Weighted On-Base Average

E. Sample Data Table

A miniature sample data table is given below in Table 1 to provide a visualization of how the data is formatted. This sample shows a condensed form of the dataset across three observations and six attributes.

F. Data Quality Considerations

There are limited data quality issues in the dataset to address. The dataset is restricted to qualified batters, excluding part-time or bench players, which favors the analysis toward established, full-time players. Additionally, because all data is

TABLE I
Sample Data

Player	Year	K%	BB%	wOBA	xwOBA
Tucker, Kyle	2021	15.9	9.3	.383	.394
Olson, Matt	2025	24.3	12.6	.366	.360
Swanson, Dansby	2024	24.3	9.1	.307	.324

taken at a season level, any in-season variation in individual plate approach and performance is masked. The dataset also does not account for the opposing pitcher, varying in-game situational plate appearances, or stadium/field differences, which may influence plate approach and outcome statistics.

III. Discovery Questions

A. Questions

1) Are there distinct, natural groupings of MLB batters based on their plate approach statistics? : This is the key question that will be answered: finding the patterns arising from how batters approach the plate and the groupings that form from them.

2) How do observed result statistics (wOBA) and expected result statistics (xwOBA) differ across these groupings? : This question examines how the different, discovered plate approach profiles are related to offensive results, both in what is observed and in what is expected based strictly on the quality of contact.

3) Are there outlier player approaches and performances that significantly deviate from what is expected from their plate approach among the groups? : This question will help identify unusual observations that feature offensive approaches and performances that do not align with the typical results seen from their respective plate approach grouping.

B. Value

Overall, these discovery questions are valuable because they reveal broad plate approach profiles among MLB batters. Furthermore, these profiles can be related to performance outcomes and variance when analyzing observed and expected outcome metrics. Finding outliers enhances the analysis by uncovering pattern exceptions, which offer insight into which plate approaches differ significantly or tend to result in over- or underperformance when compared to expectations.

IV. Implemented Data Mining Techniques

A. Clustering

K-Means Clustering was employed to determine whether distinct, natural groupings of MLB batters exist based on their plate approach statistics. Clustering directly assesses whether common plate approach profiles arise from the data by grouping player seasons based on swing rates, contact rates, and discipline measures. The method also verifies the validity of spherical data types and ensures that the patterns found within them are concrete.

B. Anomaly Detection

Anomaly detection was primarily performed using distance-based techniques to achieve two goals. Firstly, Euclidean distances from each data point to its cluster centroid were calculated to identify players who drift furthest from their assigned cluster. These are edge-case hitters who do not align exactly with one plate approach but instead adopt a hybrid strategy, or batters who showcase extreme instances of their plate approach and favor towards one-dimensionality. Then, anomaly detection was used to identify player-seasons that greatly over- or underperformed their expected outcomes compared to other batters within their plate approach profile grouping. Examining differences between wOBA and xwOBA statistics through Z-Scores within clusters directly located this unusual behavior, indicating disproportionate skill or variance in a batter, or just an unlikely occurrence of a plate approach profile deviating from typical patterns. The first anomaly detection implementation focuses on integrated plate approaches, while the second concentrates on irregular performance divergence.

V. Analysis Pipeline

Fig. 1 depicts the data analysis pipeline of the project. Data inspection and preprocessing consisted of collecting, cleaning, and finalizing the dataset. Exploratory data analysis was conducted using detailed univariate, bivariate, and multivariate analyses. Feature engineering, selection, and transformation finalized the features to be used in clustering and ensured the data was transformed in a way that properly facilitates clustering algorithms. Finally, the data mining stage applied the techniques described in Section IV to the data to highlight its patterns and characteristics, whilst also drawing interpretative value from them.

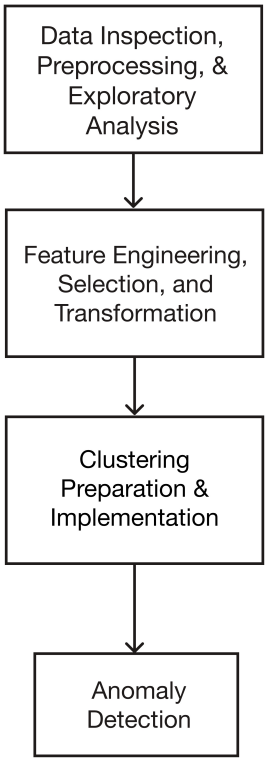

Fig. 1. Pipeline Flowchart

VI. Exploratory Data Analysis

A. Data Inspection & Preprocessing

The preparatory inspection and preprocessing of the dataset found no critical errors. Each row and column held valid values and were properly filled in, and no duplicate values were encountered in the data. The preliminary distributions of each feature were normal and standard in terms of central tendency and overall dispersion.

B. Univariate Analysis

The univariate analyses of each feature yielded results similar to one another. The distributions exhibited a unimodal distribution on their respective scales, displayed through histograms and box plots. Variations included right skews, left skews, or neither, revealing several outliers at both ends of some graphs, but nothing much beyond that. In short, these examinations produced no prominent results, paving the way for feature comparison to uncover the intricacies of the dataset as a whole.

C. Bivariate and Multivariate Analysis

When comparing these features through bivariate and multivariate analyses, direct and inverse relationships appeared. Strong correlations arose between the different wOBA statistics, contact percentages, swing rates, and strikeouts. A correlation heatmap revealed these patterns, and subsequent scatterplots were formed to confirm them. Together, they showed that the statistics grow in meaning when combined in groups, establishing the need for a diverse set of features to demonstrate key relationships in the dataset.

VII. Feature Engineering, Selection, & Transformation

A. Feature Engineering & Selection

The dataset features needed modifications before beginning the data mining phase. Firstly, a performance difference statistic was created to measure the difference between a player's wOBA and their xwOBA. This data point provides the essential information to view whether a player under- or overperformed in a given season, without taking into account the strength of their recorded performance outcome. This is ideal for identifying clusters with better or worse expected numbers to determine the stronger plate approaches better. Additionally, the Swing% characteristic was removed. As found in the bivariate analysis, this measurement served only as a mean between Z-Swing% and O-Swing%, rendering its purpose redundant. Therefore, it was omitted to clarify cluster creation and produce clearer relationship evaluations. All other features were kept as is.

B. Data Transformation

The data was transformed for clustering through standardization. Each feature was on a percentage scale but was not properly aligned within its range, which would break future algorithms. Standardization solved this problem by making each range proportional to the others for proper comparisons, retaining variability and outlier handling while centering each around zero with consistent standard deviations.

VIII. Clustering

A. Algorithm Selection

K-Means was chosen for cluster creation because the dataset was found to be normally distributed, making it well-suited to spherical cluster centers that represent precise and meaningful batter profiles.

B. Number of Clusters

The process of choosing the number of clusters to extract from the dataset was performed using the elbow method and by taking the corresponding silhouette scores. Based on the elbow method, a value of k = 3 was selected as the optimal value for maximizing in-cluster variance while minimizing the total number of clusters. The most competitive silhouette scores came from k values 2 and 3, with the former holding a marginal edge. Despite this, a total of three clusters to pull from the dataset was kept to make for a more diverse group and broaden relationship understanding. This decision retains knowledge without sacrificing dataset complexity, key attributes, or oversimplification, resulting in a more comprehensive layout for in-depth analysis.

C. Cluster Characterization

Clustering implementation found three distinct clusters. A grouped bar chart and varied scatterplots indicated these sections of clusters and visualized them distinctly.

1) Cluster 1: Cluster 1 ('Selective Batters', 196 points) holds hitters who show patience at the plate. They have the lowest swing and contact rates but the highest walk and strikeout rates, displaying extreme plate discipline, for better or worse. Their performance is also significantly the best of the clusters, and they notably underperform their expected statistics as well. In theory, this makes them the best plate approach in terms of average offensive value.

2) Cluster 2: Cluster 2 ('Swing-Heavy Batters', 251 points) holds hitters who demonstrate lots of plate aggression. They hold the highest swing rates at the price of high strikeout and low walk rates. While some batters can perform very well by using this strategy, many can also suffer from it, defining the approach as a double-edged sword. Their performance falls much more in line with the mean wOBA, and they underperform that number by a very small amount.

3) Cluster 3: Cluster 3 ('Contact Batters', 222 points) holds hitters who make well-above-average contact with balls, holding the clear best numbers in this category. They also strike out the fewest between the three clusters. Evidently, their utility comes from putting the ball in play, opening up opportunities on the field, and leaving more room for luck than usual. Performance-wise, they are on par with Cluster 2 but slightly overperform their expected statistics.

D. Clustering Findings & Interpretation

The first two proposed discovery questions can now be answered. There are distinct groupings of MLB batters based on their plate approach: aggression-based, contact-based, and swing-heavy. They all feature unique characteristics and lead in the approach measurements for their specific specialties. Each approach exhibits its own patterns and territory, cementing cluster clarity. Observed result statistics differentiate most within Cluster 1, which marks a considerable advantage over the other two clusters. Their offensive value is generally the most effective. Expected result statistics are also most strongly pronounced in Cluster 1, which largely underperforms from what is expected. The other clusters only very slightly over- or underperform, making this distinction notable.

Overall, the clustering findings reveal both surprising and expected results: plate discipline does not necessarily translate into fewer strikeouts, strikeout rates do not affect wOBA much, and selective batters are the most productive. This last point, however, has limitations. The sample size for Cluster 1 is the smallest of the three clusters, and the players in this category may be treated differently from others. This can be seen in varying batting order positions, pinch-hit opportunities, and roster constructions. Further and deeper analysis is required to validate the preliminary trend found. Most importantly of these points, however, is that players can relate to any cluster and still perform well or poorly.

IX. Anomaly Detection

Anomaly detection was broken down into locating structural anomalies, based purely on batter plate approach within each cluster, and performance anomalies, based purely on outcome results for batters comprehensively, in order to expand on the results found in the clustering phase.

A. Structural Anomalies

To find deviations within the structures of each cluster, Euclidean distances were taken between each data point and its respective cluster centroid. The top 5% of distances for each cluster were identified as anomalies due to their unusualness, without being overly strict. Additionally, Euclidean distances were drawn to the other cluster centroids, allowing differences to be measured between the closest cluster and the next closest one. The lowest 5% of these calculations were similarly gathered as anomalies. These two sets of anomalies were labeled as outlier anomalies and hybrid anomalies, where outliers represented players far from their cluster centroid, and hybrids represented players at the boundary between two clusters.

Outliers were identified as instances of extreme plate approaches within each cluster. These batters essentially doubled down on the aspects of their plate approaches that set them apart within their cluster, exemplifying exaggerated, unusual profiles, with large ranges across attributes within each cluster, indicating extreme and scattered spreads on all fronts. The resulting exhibitions featured unusually high strikeout rates for outliers in Cluster 2, for instance. Each cluster contained nearly equal numbers of these anomalies. Hybrid players were players holding a mixture of two plate approaches, with cluster distributions identical to those of outliers. Their split location between two cluster centroids indicated this, as seen by comparing their attribute ranges to those of outliers: hybrids exhibited values more tightly packed and similar, the opposite of what outliers produced. Hybrid players demonstrate prominent features across multiple clusters, but the clustering phase forced a single assignment, leading to a misrepresentation of these batters.

B. Performance Anomalies

Major performance anomalies within each cluster were identified by standardizing each data point's performance distance metrics using Z-scores to find the most unusual cases of over- and underperformance based on differences between wOBA and xwOBA. Each data point more than 2 standard deviations from the mean was defined as unusual, with positive values indicating overperformance and negative values underperformance.

More occurrences of overperformance than underperformance were observed, with a difference of six. Underperformance instances were evenly distributed across the clusters, while overperformance was not: Cluster 3 was skewed, with ten instances, and Cluster 2 was underrepresented, with four instances. Aside from cluster denotations, over- and underperformance occurred at random across their analysis and visualizations, and overperformers showed greater variability in contact rates. This aligns with the wOBA statistic, which accounts for factors such as luck and in-game external factors, whereas xwOBA only assesses based on the quality of contact made on balls.

C. Anomaly Detection Findings & Interpretation

The final proposed discovery question can be answered with the results of the comprehensive anomaly detection conducted.

Regarding diverging plate approaches, outlier and hybrid players emerged, each representing unique changes in their plate approaches relative to their cluster grouping. There is no single cluster that is more susceptible to these anomalies; they appear to be natural, balanced phenomena that occur across the board. Additionally, they clarify hitter approaches, serving as an extension of clustering, which simply captured general tendencies. Outliers showcase the span of individual plate approaches, showing the different ways in which a player can commit to one. Hybrids define multiple plate approaches and live in the overlaps of clusters, indicating that not every player fits perfectly under one umbrella in the way that K-Means clustering imposes. Outlier players are more static and predictable because they are radically specialized in their approach, while hybrid players are more dynamic and adaptable, as they can utilize skills from different plate approaches in a versatile way.

The performance anomalies found were more volatile. Contact hitters dominated overperforming batters, signifying that contact is the one controllable parameter in the wOBA

calculation. The high variance of the attribute means that, generally, more contact is associated with more favorable luck. Conversely, limited contact from Cluster 2 batters leads to the opposite, which is why these batters are less prominent in this feature. Underperforming players were more consistent across the data, with equal cluster distributions. This may mean that bad luck is fixed across the league and uncontrollable, while good luck can be slightly altered. The key feature of these performance anomalies is that they are, nonetheless, mostly random, with many similar players producing wildly different results at times, a result of the variability. Performance results can then be deduced as not only assembled from skill, but also a mix of randomness that cannot be predicted, all of which is independent of batter plate approach.

Very few of the anomalies coincided with each other. This suggests a lack of connectivity between them, meaning that none of them affects any others. Unnatural plate approaches will still mainly yield normal performance results, whereas normal plate approaches outlined by the clusters can randomly produce unexpected performance results.

X. Timeline

All that remains for the project is polishing and cleaning it up for the final deliverable. All of the code will be reviewed, optimized, and confirmed to ensure clarity and accuracy. Descriptions and explanations of techniques, analysis, and visualizations will also be refined and enhanced to provide a clear rationale for every major decision, enabling a complete understanding with no gaps. For the presentation, companion slides will be developed to summarize the entire pipeline in detail, including all key components, points, and conclusions. The final project report will be included in an organized, descriptive, and fully concluded GitHub repository.

XI. Conclusion

Exploratory data analysis, feature transformation, clustering, and anomaly detection have laid out useful patterns in individual MLB batter data that are important to current players. Players share distinct plate approaches, with varying performance outcomes, across contact-based, swing-based, and discipline-based strategies. Discipline-based batters have the highest average offensive value. Anomalies describe exceptions to these findings through the presence of outlier and hybrid players who lean toward opposite sides of the clusters. Performance unusualities also appear through over- and underperformance, moderately impacted by contact rates but mostly influenced by external luck factors. The data mining results comprehensively outline various batting archetypes across MLB, gathering player information that can be put to use across the league. Some of these real-world applications include player acquisition decisions, roster management, lineup construction, and even extend to widespread platforms such as fantasy baseball fan competitions. These and similar purposes demonstrate the need for critical batting data to accurately evaluate players without bias.

References

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